

Manual

**e-Kanban
BDE-KBN
HYDRA 8.2**

Version 1.2.23049

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1 Overview: e-Kanban

Usage

The kanban control as decentralized production control managed by staff has been designed to apply the pull principle to achieve efficient logistical structures for the production of customized goods and services.

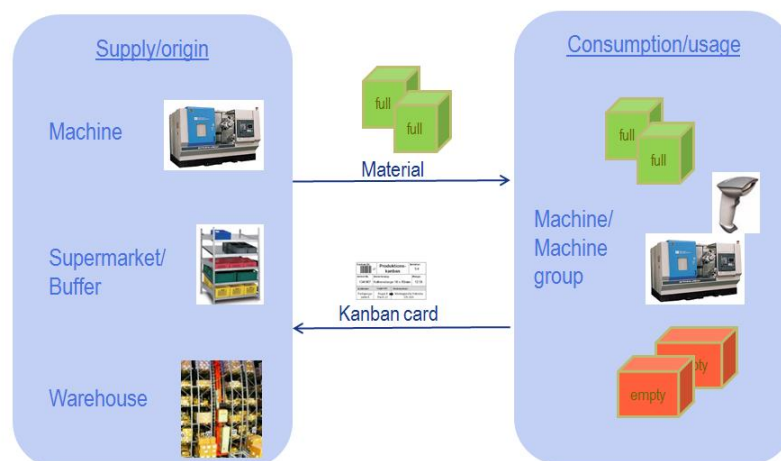
Higher-ranking structures get a new role and are re-defined based on the overall context.

Mandatory prerequisites for implementing a kanban control affecting the relevant items:

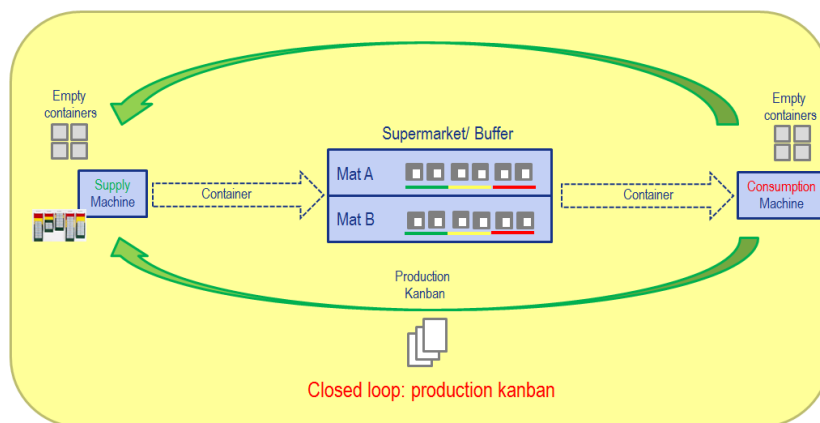
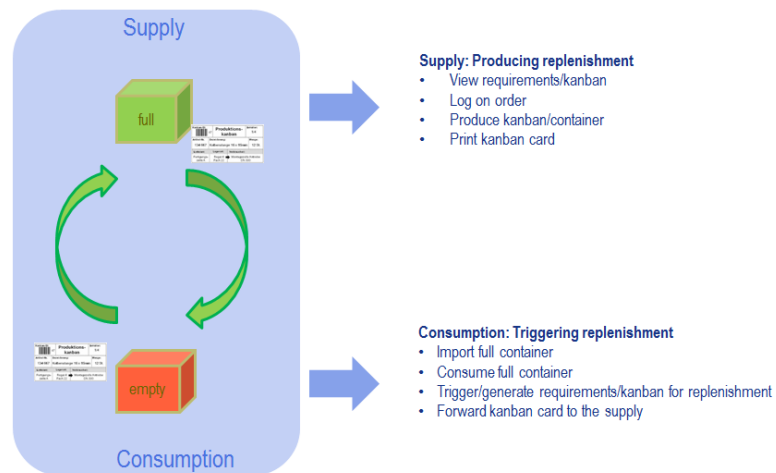
- Uniform customer demand for the relevant item
- Limited variants of the product
- Production structure based on material flow
- Short setup times

The manufacturing of an item/material is generally triggered by withdrawing a kanban container at the consumption.

Then the kanban card is transferred to the supply (e.g. preceding machine, machine group, material buffer), which in turn triggers new supplies/replenishment.



The kanban control based on the pull principle is represented in electronic form within MES and, as a result, the flow of the relevant kanban objects is displayed within their closed loop (from the supply to the consumption and vice versa).



2 Configuration eKanban

Basic configuration

If you want to install the e-kanban functions on an already existing HYDRA 8 system, proceed as described below:

1. Activate the patch dbp_mpl_ekanban as follows:
 - a. UNIX systems (run in the server prompt of the HYDRA directory):
`hydsr.out db_sql/dbp_mpl_ekanban.hsc`
 - b. Windows systems (run in a DOS window of the HYDRA directory):
`hydsr.exe db_sql/dbp_mpl_ekanban.hsc`
2. Check the patch output
3. Back up the existing dialog configuration:
 - a. UNIX systems:
`hydlgcfg.out DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=%
DATEI=dialog.dlg`
 - b. Windows systems:
`hydlgcfg.exe DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=%
DATEI=dialog.dlg`
4. Now load the new dialog configurations with the following command:
 - a. UNIX systems:
`hymw.out -u9999 -b db_sql/aip_kbn.dlg`
 - b. Windows systems:
`hymw.exe -u9999 -b db_sql/aip_kbn.dlg`

The dialogs KBN_BOARD, KBN_FILL, KBN_EMPTY are then imported **as template** (with type "AIPDEF" and dialog user "999") or existing dialogs are updated, if required.
5. Go to the MOC application: System administration → Terminals → Dynamic dialogs (transaction code ddconf). Copy the dialogs KBN_BOARD, KBN_FILL, KBN_EMPTY from the template (type "AIPDEF" with dialog user "999") to type "AIPDEF" and dialog user "0". To do so, switch to the HYDRA Professional Mode.
6. Activate the new dynamic dialogs:
 - a. UNIX systems:
`hydialog.scr AIPTNR 0`
 - b. Windows systems:
`sh.exe hydialog.scr AIPTNR 0`



This command activates the default dialogs. If you use terminal-specific dialogs in the system, these dialogs need to be modified by an MPDV consultant.

Configuration: Order types

Menu

Master data → Order → Order types

Configure one [Order type](#) for the kanban orders and one order type for the kanban capacity orders.

Kanban orders

You can use the preconfigured order type "KBN" as order type for kanban orders.

Kanban capacity orders

If you want to generate kanban orders from kanban capacity orders, you must create a separate order type (e.g. "ZKABA"). The behavior of the order type for kanban capacity orders must be similar to the behavior of production orders (order type 0).



Do not mix up the order type for kanban capacity orders (e.g. "ZKAPA") with the preconfigured order type 3.

Configuration: Order status texts

Menu

Master data → Order → Order status texts

Create a status text for planned kanban orders and operations:

Parameter name	Value
Status text	e.g. 901
Status text name	planned

Configuration: Order status

Menu

Master data → Order → Order status assignment

Configure the [status](#) "P" as order status "planned" for the order type, which is defined as order type for kanban orders (e.g. "KBN").

Parameter name	Value
Order type	e.g. "KBN"
Status	"P"

Parameter name	Value
Entry: Control	S = None
Entry: can be logged on	☒
Planning: Planning	N = No planning
Options: Initial status during creation	☒

Configuration: Operation status

Menu	Master data → Order → Order status assignment
------	---

Configure the [status](#) "P" as order status "planned" for the order type, which is defined as order type for kanban orders (e.g. "KBN").

Parameter name	Value
Order type	e.g. "KBN"
Status	"P"
Entry: Control	S = None
Entry: can be logged on	☒
Entry: Sequencing list	☒
Planning: Planning	N = No planning
Options: Initial status during creation	☒

INI configuration

Menu	System administration → System settings → INI configuration System administration → System settings → INI data configuration
------	---

Configure in the [INI configuration](#) under the name "MPL" in the "KANBAN" section:

Key	Value (example)	Comment
KBN_AUART	KBN	Kanban order type
KAPA_AUART	ZKAPA	Capacity order type



If the INI configuration KAPA_AUART is enabled, the system searches for kanban capacity orders when kanban orders are generated. Work plans are no longer used then.

Configuration: Scheduler

Menu	System administration → System settings → Scheduler
------	---

Make the following entry in the [Scheduler](#) to automatically generate kanban orders:

Type	S – Standard
Alterable	Yes
Type	I – Interval
Visible	Visible
Active	☒ Active
HYDRA user	0
Command (Windows)	sh.exe ./kbn_sgen.scr
Command (Unix)	kbn_sgen.scr
Comment	Generation of kanban orders
Interval	Hour: 1 Minute: 00

Configuration: Resource status configuration

Menu	Master data → Resources → Resource status
------	---

Configure the following three kanban [resource statuses](#) for the kanban objects (resource type):

Resource key	type	Status (example)	Designation	
KBN		1	Initial	
KBN		102	Full	
KBN		103	Empty	

Configuration: Advanced object configuration

Menu System administration → System settings → Advanced object configuration

Specify in the [Advanced object configuration](#) which status is used:

Parameter name	Value
Object type	MPL
Object ID 1	Initial status, e.g. 1
Object ID 2	Status for filled kanban, e.g. 102
Object ID 3	Status for empty kanban, e.g. 103
Parameter	RESTYP
Parameter value	KBN

Configuration: Number ranges

Menu System administration → System settings → Number ranges

Create a [number range](#) for the kanban orders. This number range is used for the generation of kanban orders.

Parameter name	Value
Object	AUNR
Key	AART
Value	KBN (value from INI configuration for KBN_AUART)
Assignment code	NUM
Prefix	KBN (see below)
Range from	100000 (see below)
Range to	199999 (see below)
Current value	0



The number of digits resulting from <prefix> and range from/to must be identical to the order number length configured in the [Basic settings](#).

AIP configuration



Minimum AIP version for e-kanban: 2.0.2.80

Display of electronic kanban board

Configuration in ctaipplay.ini (AIP 8.1, AIP 8.2):

```
[WF@KBOARD]
CMD=DLG=LIST;u_l_kbn|MOD=B|KBN.TNR:ID=<*TNR>|
FILTER=
SECTION=Kanban table
DATAFIELDS=
FILE=kanban.lst
AUTOFILTERCOL=
KBN_INTERVAL=60
;ORDER=KBN.STA=-

[Kanban table]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

GRID_CELLPAINT=ON
EXAMINE_CELLBKCOLOR=STA1,KBN.STA,1-clLime|2-clYellow|3-clRed

;Status
ALIAS STA1=(DUMMY)=C1,50,L,Status
KBN.STA=C1,0,L
;Kanban material number
KBN.ATK=C25,145,Z
;Control loop
KBN.ID=C15,90,Z
;Withdraw
KBN.ZLO=C30,90,Z
;Total quantity of kanban objects in circulation
KBN.ANZ=N12,90,R
;Configured/ planned quantity of the kanban object
KBN.EGR:P=N10,90,R
;Full KBN
KBN.ANZ:F=N10,90,R
;Empty KBN
KBN.ANZ:E=N10,90,R
;Minimum stock level
KBN.MIN=N10,100,R
;Maximum stock level
KBN.MAX=N10,100,R
```

Extensions of the order list

You can add configurations in ctaipplay.ini, section "[sequencing list (auto)]" (AIP 8.1, AIP 8.2):

```
; Enable coloring specific to the cell:
GRID_CELLPAINT=ON
; KANBAN orders: orange (here AUART=KBN - can deviate)
EXAMINE_CELLBKCOLOR=COL1,AUART,KBN-$0080FF
; Foreground color: planned/prepared OPs
EXAMINE_SCANEXPR1=AST=P
EXAMINE_SCANCOLOR1=clGreen
EXAMINE_SCANEXPR2=AST=V
EXAMINE_SCANCOLOR2=clBlue
; Dummy column for orange coloring
ALIAS COL1=(DUMMY2)=C1,100,L
; required column as color reference (is not shown)
AUART=C1,0,L
```

Fill kanban

Configuration of list in ctaipplay.ini (AIP 8.1, AIP 8.2):

```
[control loops]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_CAPTION=control loops

;KBN.STA= C15,120,L
;KBN.ATK=C15,120,L
KBN.ID=C15,120,L
KBN.ZLO=C15,120,L
KBN.PUFFER=C15,120,L
KBN.ANZ=N4,80,R
KBN.MENGE=N4,80,R
KBN.ANZ:F=N4,80,R
KBN.ANZ:E=N4,80,R
KBN.MIN=N4,80,R
KBN.MAX=N4,80,R
KBN.EGR:P=N4,80,R

[Kanban objects]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_CAPTION=control loops

;KBN.STA= C15,120,L
;KBN.ATK=C15,120,L
;KBN.ID=C15,120,L
KBN.RES:NR=C15,120,L
```

```
KBN.RES:LNR=N4,80,R  
KBN.ZLO=C15,120,L  
KBN.PUFFER=C15,120,L  
KBN.ANZ=N4,80,R  
KBN.MENGE=N4,80,R  
KBN.ANZ:F=N4,80,R  
KBN.ANZ:E=N4,80,R  
KBN.MIN=N4,80,R  
KBN.MAX=N4,80,R  
KBN.EGR:P=N4,80,R
```

Configuration of keys in ctaipbut.ini (AIP 8.1):

```
<n1>=KBN_BOARD,electronic kanban board  
<n2>=KBN_FILL,fill KANBAN
```

with <nx>=1, 2, 3, ... - depending on where the buttons are placed.

Configuration of keys in I_main.xml (AIP 8.2)

- Create a backup of the file I_main.xml.
- Open the file I_main.xml using a text editor.
- Search for the text
`<Caption Function="Translate" LanguageKey="lkeKanban">eKanban</Caption>`
- In the row above, change the value from false to true:
`<Visible>True</Visible>`
- Save the file I_main.xml and restart the terminal.

3 e-Kanban

Utilization

The kanban control as decentralized production control has been designed to apply the pull principle.

MES represents kanban control based on this principle in electronic form (e-kanban). Therefore, the flow of the relevant kanban objects is displayed within their closed loop (from supply to consumption and vice versa).

Prerequisites

These general definitions need to be taken into account for using e-kanban:

- Instead of physic kanban cards, MES uses or manages (electronic) kanban objects.
- The ideal total quantity of circulating kanban objects has to be calculated in advance. MES does not provide calculations or algorithms for this purpose.
- In MES the kanban objects are always posted from one material buffer to the next. It is not intended to repost them to the object "machine" or "machine group".
Consequently, supply and consumption are represented as material buffers in the system and to be configured accordingly.
- The basic package first only refers to the simple case. Here, the closed loop between one supply and one consumption is only taken into account, i.e. it is not taken into account how many machines actually pertain to the consumption and how often the material/kanban object physically moves from one "consumption buffer" to the consuming machine.

The kanban objects are not traced within the supply even if it comes to multi-level processes (kanban order includes several OPs) for replenishment production.

Procedure

Creating kanban objects

The kanban objects are created and managed in MES. Generally, a status is kept for the objects. Once the kanban object has been created, it is assigned the status "initial".

Generation of filled kanban object at the supply

The supply (material buffer) for replenishment production can be operated by one or several machines.

A machine produces the relevant material/article for a kanban order.

If a kanban object is filled, the status of the kanban object is set to "filled" by the dialog "fill kanban". However, it is irrelevant at which machine of the supply the kanban object was produced.

The kanban object is directly posted in the material buffer of the consumption defined within the closed loop.

Provision of the kanban object at the end point: consumption

The kanban object is provided at the consumption and, once it has been filled, it is directly posted into the material buffer of the consumption that is defined within the closed loop.

In MES the consumption (material buffer) is represented as:

- A general material buffer (supermarket for several machines/machine groups). In this case, the system cannot trace the kanban object (to which machine it was sent)
- The material buffer of the machine (preceding)

Emptying the kanban object at the end point: consumption

The kanban material provided at the consumption is "used/consumed" for production at a machine.

If the kanban material of a container is completely consumed, the kanban object is set to the status "empty" by the dialog "empty kanban" and directly posted onto the defined material buffer of the supply.

However, it is irrelevant to the kanban process at which machine following the consumption the container was actually emptied.

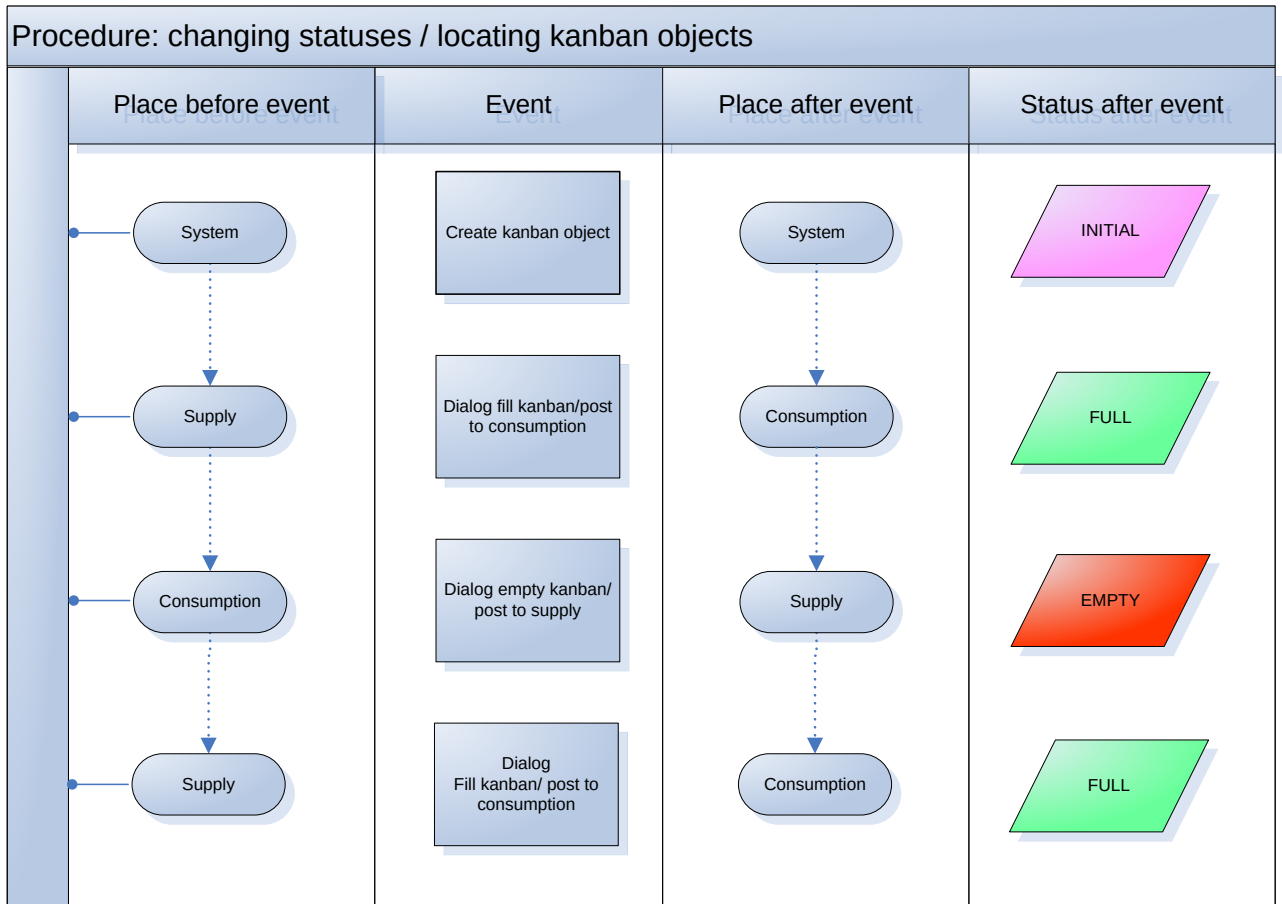
Generation of orders on request

If required, the consumption takes the kanban object from the supply. If machines are indirectly regarded as subordinate elements of the supply, replenishment production will be triggered by consuming or emptying the kanban objects (dialog: empty kanban) at the relevant machine/terminal.

The requirement of a new kanban object for this material/closed loop is visualized immediately at the supply/terminal - electronic kanban board.

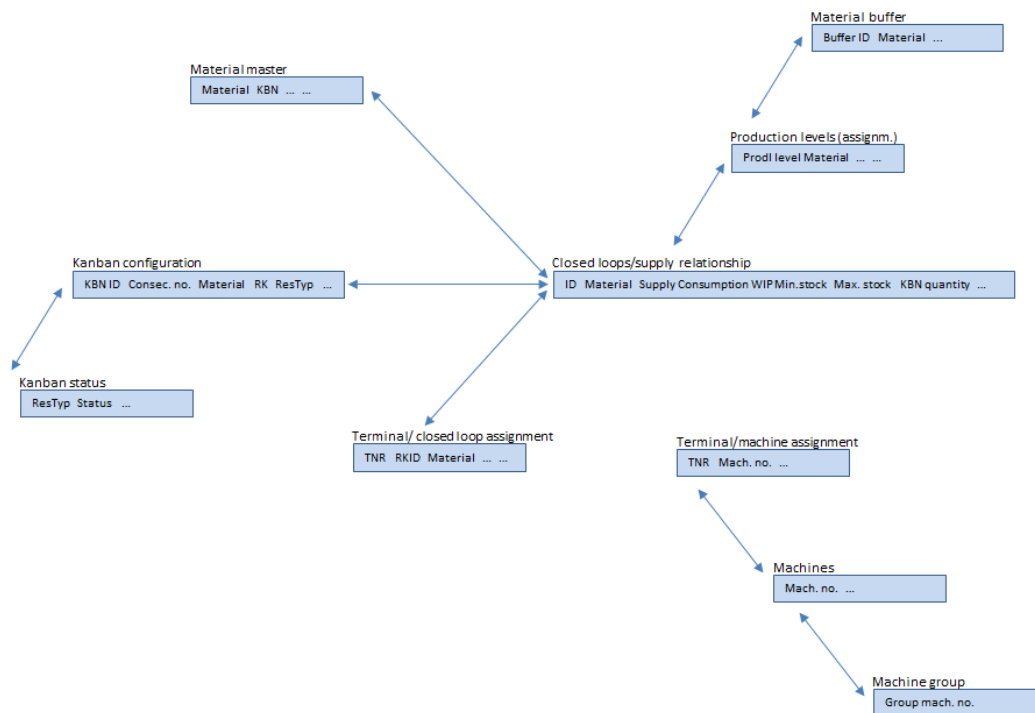
Once a minimum quantity of empty kanban objects has been reached, a planned kanban order that is shown within the order list at the supply can be generated automatically. The planned kanban order is enhanced by any further empty kanban object.

If the maximum quantity of empty kanban objects is reached, the planned kanban order becomes a fixed kanban order that can no longer be changed. Quantities are no longer updated. The fixed kanban order is also shown in the order list at the source.



Used master data and their connection

The following, cooperating master data are used to represent the kanban process within MES. They need to be created in the system.



Order generation

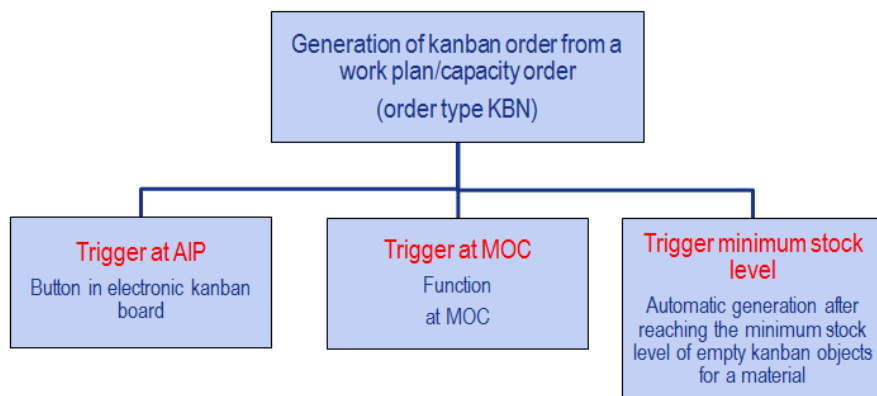
General:

In general, a kanban order is generated from a work plan created in MES or from a capacity order provided by the ERP system for the order type "KBN" (can be named flexibly). However, the generation itself might be due to several different situations.

Consequently, the following options to generate kanban orders are provided:

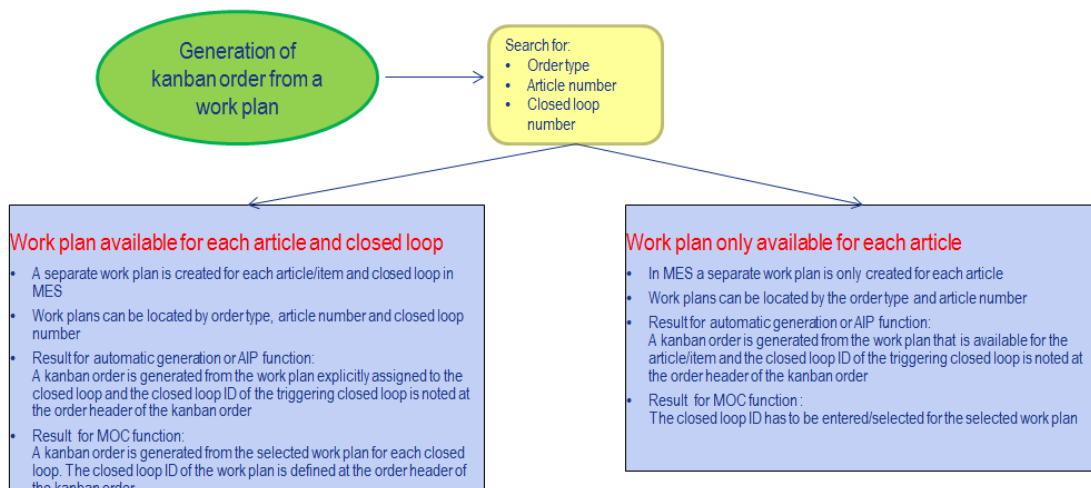
- **Manually at the AIP**
The dialog "electronic kanban board" provides the option to generate a kanban order for an article/material and closed loop at the push of a button.
- **Manually at MOC (function of the application "kanban stock")**
Orders pertaining to kanban orders are generated manually by the foreman at MOC from a separately created work plan or a provided capacity order at least for each article or article and closed loop.
- **Automatic generation**
The system automatically generates a planned kanban order for the relevant article from a work plan or provided capacity order when the "yellow zone" (minimum quantity of empty kanban cards) has been reached.

- Quantities are updated for the kanban order by every new, empty kanban object.
- The worker may start the generated, planned kanban order at the AIP at any time.
- To generate kanban orders, at least one separate work plan is created or capacity order is transferred from ERP for each article.
- Once the "red" zone (allowed maximum quantity of empty kanban cards) has been reached, the quantities are updated for the planned kanban order the last time and the planned kanban order becomes a fixed kanban order.

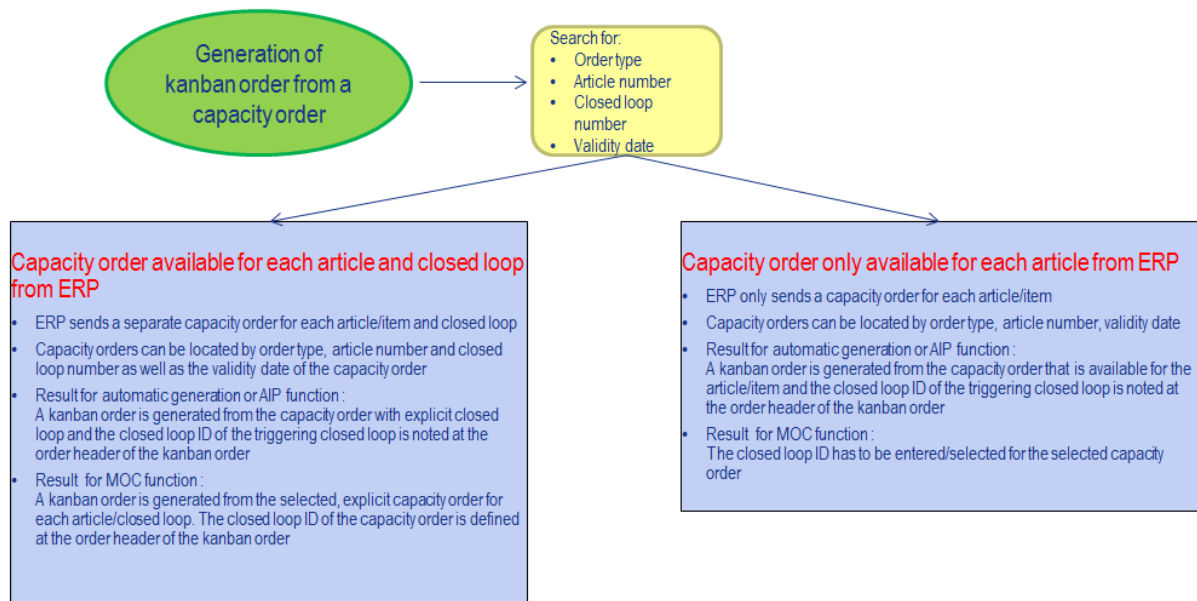


The work plans to generate kanban orders are created in MES or capacity orders are transferred from ERP. The system considers the below-mentioned situations:

Generation from a work plan:



Generation from a capacity order:



Utilization of a work plan/capacity order

- Utilization of a work plan/capacity order **for each article and supply relationship/closed loop**

Even though there might be several supply relationships/closed loops (ways from supply to consumption) for one article/item, the system has to generate a relevant kanban order for exactly one specific closed loop. The work plan/capacity order defined for this article/closed loop is used. The ID of the supply relationship/closed loop is entered in the header data to find the correct work plan / capacity order (shown in work plan header/ order header --> assignment --> master data --> supply relationship/ closed loop).

If no closed loop ID is available at the work plan/capacity order header, the first work plan/capacity order found for the article number will be used.

The closed loop number of the **triggering** closed loop is taken over for the generated kanban order.

- Utilization of a work plan/capacity order **for each article only**
 - This is sufficient, for example, if there is only one supply and consumption for each article, i.e. there is only one closed loop and no differentiation has to be made. In this case, it is not necessary to define a closed loop number for the work plan header/capacity order header.

If no closed loop ID is available at the work plan/capacity order header, the first work plan/capacity order found for the article number will be used.

For which closed loop the generated kanban order is used, results from the assignment of the closed loop ID for which the kanban order is triggered/generated (e.g. by using the relevant AIP function or the automatic generation function). The closed loop number of the triggering closed loop is taken over to the generated kanban order.

- It might be the case that there are several closed loops for each article but the same and only created work plan/capacity order is always used. Then it is not necessary to define a closed loop number for the work plan/capacity order. For which closed loop the generated kanban order is used, once again results from the assignment of the closed loop ID for which the kanban order is triggered/generated (e.g. by the relevant AIP function or the automatic generation function).
- Another possibility is to define a work plan/capacity order for each article/item only but to enter or select a specific closed loop number when generating a kanban order by using MOC functions. For which closed loop the generated kanban order is used, results from the assignment of the entered closed loop ID for which the kanban order is triggered/generated.

Prerequisites, operating conditions



- It has to be defined in advance whether a system uses capacity orders or work plans to generate kanban orders. Both variants cannot be used at the same time. The system uses the INI configuration specifying if work plans or capacity orders are to be used.
- Within the system work plans relating to articles or to articles and closed loops generally have to be defined for the order type "kanban orders" (e.g. "KBN").
- In general, the planned kanban quantities have to be defined for all kanban articles within the system.
- Number ranges for the automatic generation of the order number for kanban orders have to be created or used within the system. Number ranges can be configured flexibly and depend on the order type.

Data:

Input data to find the appropriate work plan/capacity order to "generate orders":

- Article number of the triggering article (from AIP function, MOC function, automatism)
- Closed loop ID of the triggering closed loop (from the relevant AIP function, MOC function, automatism)
- All data defined in the capacity order or work plan

The generated kanban order uses the following data from the provided work plan or capacity order:

- Scheduled start → point in time of the generation
- Scheduled end → point in time of the generation + quantity * target cycle → standard formula
- The work plan/capacity order may include several operations; however, in general it is only one OP. They are taken over for order generation and used for planning based on the default configuration.
- The quantities for the generation depend on the number of empty KBN objects and planned KBN quantities (from kanban master data) for the triggering article/closed loop and are calculated for the kanban order.
- The machine group of the OP/OPs is entered in the work plan/capacity order and planning is based on it.
- Target data (target cycle, partitioning, bill of material, etc.) is taken over from the work plan/capacity order.
- The article numbers of the order header and the last OP are identical. The article number from the order header is used for order generation. The other article numbers of the order are filled based on the work plan/capacity order.
- The closed loop ID of the triggering closed loop is used for the kanban order. The closed loop ID that is defined for the work plan/capacity order is not used.
If the kanban order is generated by the MOC function "generate order", the entered closed loop ID is used and transferred to the kanban order.

Procedure/processing

A kanban order can be generated by the following three possibilities:

- Manually at MOC
Utilization of the function "Generation of kanban orders using MOC" and generation of a kanban order from a defined work plan or capacity order.
- Manually at AIP
The "electronic kanban board" of AIP enables the generation of kanban orders for a selected article and closed loop. The order is generated from a work plan/capacity order created for this article/martial and closed loop.
- Automatic generation
If the minimum stock level of empty kanban cards (yellow zone, e.g. ≥ 5) that is configured within the supply relationships/closed loop for an article (e.g. 5 empty kanban objects) is exceeded, a planned kanban order is generated from the work plan/capacity order defined for the article and closed loop.

Once another kanban object with the status "empty" is available, the number of empty kanban containers increases and the planned kanban order is increased by the empty kanban object (e.g. the included planned kanban quantity):

- quantities are updated for the planned kanban order by every additional empty kanban object
- the worker may start the generated, planned kanban order at the AIP at any time

the order/operation status of the kanban order remains "planned" and the OP can be logged on.

Once the allowed maximum quantity of empty kanban cards ("red" zone) has been reached, the quantities are updated for the planned kanban order the last time and the planned kanban order becomes a fixed kanban order

The order/operation status of the kanban order is prepared and the OP can be logged on.

The kanban order is no longer changed by an automatic or manual quantity update.



Please note: This function can be enabled/disabled. This function cannot be used in a reasonable manner if the generation from the electronic kanban board is to be used.

The actual logic/function for generating orders is triggered by the mentioned three possibilities.

Result:

The article number and closed loop ID of the triggering closed loop or the triggering function are used as input data for all variants to find the appropriate work plan or capacity order and to generate the kanban order.

- The order or operation quantity refers to the number of currently empty kanban objects from this closed loop times the defined kanban quantity from the supply relationships/closed loops.

Order quantity/operation quantity = number of empty kanban objects x defined kanban quantity of the material

- The order type = KBN
- The order/operation status of the kanban order is prepared and the OP can be logged on
- The closed loop ID of the triggering closed loop is transferred automatically to the order header (in assignment --> master data --> closed loop) of the generated kanban order. The closed loop ID that might be defined for the work plan/capacity order is not taken over.



- A cyclic program checks for which article and closed loop the minimum stock/maximum stock of empty kanban objects has been reached and successively generates one order after the other of the kanban order.

- While a kanban order is being generated for an article/closed loop, this entry is blocked for other functions (this prevents the AIP user from generating a kanban order for the same closed loop during automatic generation).

Return values

- General
 - Shows the generated kanban orders/OP in the status "prepared" within the MOC order overview
 - Shows the generated kanban orders/OP in the status "prepared" within the AIP sequencing list (selected as fixed kanban orders/selected as planned kanban orders)
- The generated kanban order includes an assignment for the closed loop at the order header for which it was created (see *order info* → *tab order header assignment* → *master data* → *closed loop number*).

Capacity orders/ERP interface

General:

There is a general interface connection to diverse ERP systems for e-kanban (SAP-PPDC and EIS-ERP). Consequently, orders can be transferred from ERP to HYDRA and, vice versa, uploads can be posted on them (recorded durations and quantities).

The transferred orders differ from common production orders, as they are "capacity orders". In contrast to common production orders, they have the following tasks:

- Capacity orders are provided to generate planned/fixed kanban orders within MES. Within MES the capacity order is regarded as a work plan or template to create kanban orders and, therefore, it includes all relevant data required to produce operations within the order structure.
- Consequently, capacity orders are generated within ERP and transferred to MES including the BOM, PRT list and additional data (target specifications) that are determined from ERP master data.
- Normally, capacity orders are built on a monthly basis as capacitive forecast and transferred to MES.
- The planned target quantity of the capacity order is reduced by the number of actual quantities to be produced for a generated kanban order that is planned/fixed or with each quantity update for the kanban order. This is shown in MES.

- The recorded quantities and durations of a kanban operation are also posted in relation to orders on the capacity order/operation and uploaded to ERP. Thus, goods movements or consumption can be performed in a retrograde manner in ERP.

A goods movement is not generated and transferred to ERP every time a kanban object is produced.



If work plans are already used as template to generate kanban orders, **capacity orders cannot be used at the same time.**

In MES the capacity order is **not** used for capacity planning/machine assignment in HLS.

Prerequisites, operating conditions:

- Utilization of the standard ERP interface for production orders (including component list)
 - EIS-ERP
 - EIS-EZI
- Utilization of the standard SAP interface for discrete production orders (including component list, order type)
 - SAP-PPDC
 - SAP-ISS

Data within the capacity order:

These definitions have to be made or this data has to be transferred within the order interface.

ERP → MES (Download)

- Order header data
 - Order type for capacity orders = e.g. ZKAP (Please note: the order type can be configured)
 - Article for which the kanban order is to be generated
 - Closed loop for which the kanban order is to be generated (new field in ERP-EIS and SAP-ISS/function module AK)
 - Target quantity (monthly target quantity)
- Operation data
 - Article
 - Machine group
 - Flag "planned for machine group"
 - Target quantity (monthly target quantity)

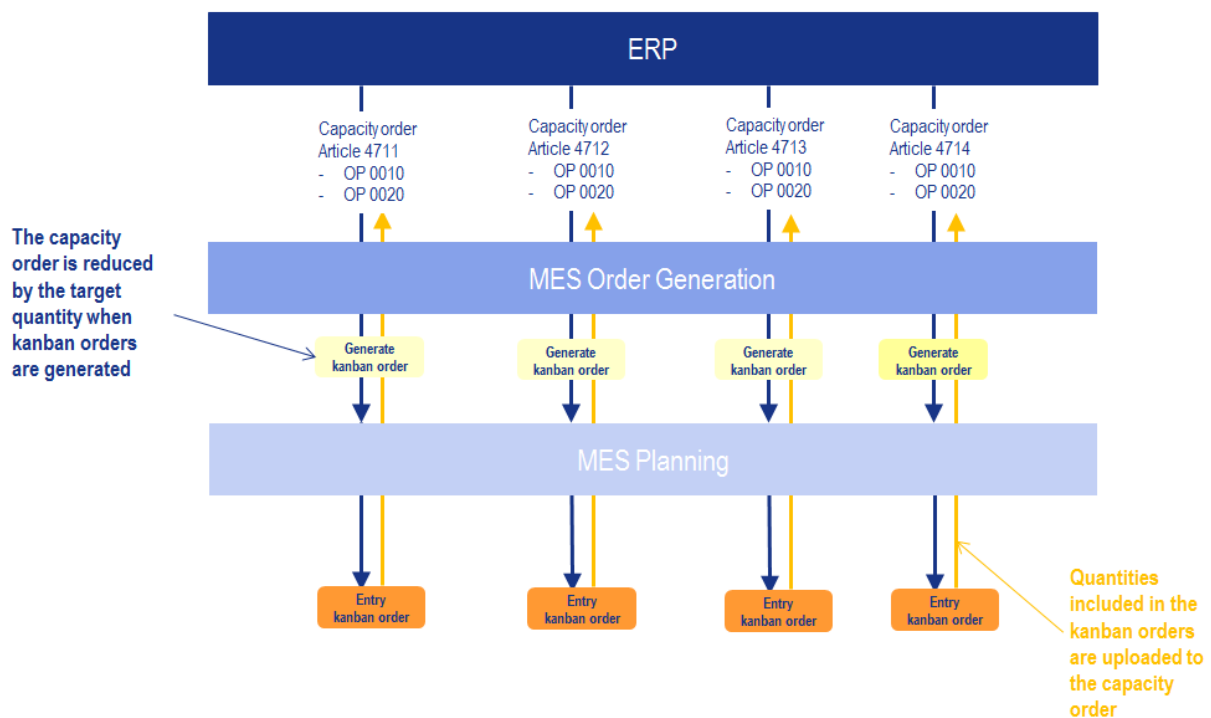
MES → ERP (uploads)

Relevant data included in uploads for capacity orders:

- Capacity order number
- Durations from kanban orders
- Quantities from kanban orders
- Please note: The machine number might not be known in ERP

Procedure/processing

Capacity orders are generated in ERP. Once they have been released, they are transferred to MES. In the first place, capacity orders are used as "templates" to generate kanban orders in MES. In general, the procedure is as follows:



Transfer of capacity orders from ERP

ERP transfers the capacity order e.g. with the order type "ZKAP" (can be configured). MES takes over the capacity order and displays it in all relevant standard applications (just as it is the case for production orders). These are, for example:

- Inbound transactions
- Backlog of orders
- Order information

In MES the capacity order is used like a work plan for an article and closed loop.

Once collected in MES, the capacity order may have the following statuses for the different objects:

- Order header → prepared
- Operations → prepared

The system may only include one valid capacity order at a time from which kanban orders can be generated. Even though several capacity orders might exist for an article/closed loop in the system.

The valid capacity order is always the newest/most recent capacity order within the system.

Utilization of capacity orders in MES

Capacity orders are used as "replacement work plan" and kanban orders are generated from them. For this reason, kanban orders are generated by using the same functions as for work plans. Triggers are:

- Manually at AIP
- Manually at MOC
- Automatic generation of orders

Generation of orders from capacity orders

Orders are always generated based on the same logic. The found capacity order (for article, closed loop, validity date) is used as template and a kanban order is generated.

The procedure for generating kanban orders from capacity orders is the same as for generating them from MES work plans.

The essential difference for the generation of kanban orders from capacity orders is that the target quantity within the capacity order is reduced by the order quantity of the kanban order every time a kanban order is generated.

With the automatic generation of planned/fixed kanban orders, the target quantity within the kanban order is increased and the target quantity within the capacity order is reduced every time quantities are updated (with every new, empty kanban object for this article and closed loop).

Uploads to capacity orders

The produced quantities and durations from the kanban order are transferred to the capacity order (added up).

For capacity orders, uploads relating to orders are also transferred to the ERP system. They are generated if a kanban order is interrupted/finished.

Status of a capacity order

The capacity order may have the following order/operation statuses (they are identical):

- Prepared: The capacity order has been transferred from ERP and created in MES.
- Started: The first kanban order has been generated for the capacity order.
- Finished: the last kanban order has been generated for the capacity order and the capacity order is no longer applicable (a newer capacity order is available).

4 Material Master

Overview

Menu	Master data → Material → Material master
Transaction code	matc
Function authorization	matc

Available user fields		
Where?	Object type/user field key	Source (type)
Detail view	ARTIKEL/SYSTEM	Material (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

You use this function to create a material master in the system for the materials used.

Integration

The material master has been designed to edit materials. This refers to especially defined master data. The material data defined in the master data are used by the composition function or e-kanban systems.

Requirements

When you create the master data for a material, the material type and the material buffer must already exist in the system.

Selection criteria

The application provides the following selection criteria:

Tab *General*

Material number

Material number

Drawing issue number

Drawing issue number of the material, also referred to as index

Designation (name)

Material name

Inactive

Inactive, active materials. The checkbox is not enabled by default.

Tab *Composition***Scrap material**

Selects materials identified as scrap material.

End product

Selects materials identified as end products.

Input material

Selects materials identified as input material. The checkbox is enabled by default.

Tab *Kanban*

To show these fields, you require the user authorization "kov".

Kanban material

Selects the materials used in the kanban process.

Tab *User fields*

You can use this tab to perform a selection according to the defined user fields for object type "ARTIKEL" and the user field key "SYSTEM".

Field descriptions**Material number**

Material number

Drawing issue number

Drawing issue number of the material, also referred to as index

Designation (name)

Material name

Material type

Material type of the material

Specific weight

The specific weight of the material in the unit g/mm³

Inactive

Inactive, active materials. The checkbox is not enabled by default.

Input material

The material is an input material of the composition. This option has to be enabled for materials used in composition.

Scrap material

Identifies a material as scrap material.

Material buffer

Material buffer of the material

Fragmented size

The material piece size in kg.

Price

The price of the material in €/kg

End product

Identifies a material as end product of the composition.

Tab Kanban

To show these fields, you require the user authorization "kov".

Kanban material

Identifies a material as kanban material.

Tab MSL (only with HYDRA for Electronics)

The Moisture Sensitivity Level (MSL) specifies the moisture sensitivity of semiconductor components during packing, storage and assembly.

MSL material

This option specifies if the MSL level is relevant to material of this material type.

MSL term

Time period in hours in which the material must be used after opening.

Level/threshold

Specifies the level of the MSL material.

Comment

You can enter a comment in this field.

Tab *User fields*

You can use user fields to store additional customer-specific information in the MES. The *user fields* tab includes eight sub-index tabs, which each has eight additional user fields. The so-called user field key specifies the available user fields and their meaning.

Object type

Default "ARTIKEL"

User field key

Default "SYSTEM"

User fields

The following user fields are available after system customization (maximum):

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

By default, no user field keys are defined. You must activate in the system via customization that this kind of user fields is supported.

Toolbar



Composition recipe

This function calls the relevant composition recipe for the selected material.

5 Production Levels

1.1 Summary

Menu	Master data → Material → Production Levels
Transaction code	plev
Function authorization	plev

Utilization

The function is used to create or modify production levels within the system and to assign machines or material buffers for each production level.

Integration

The creation of production levels allows for machines and [material buffers](#), which may be used in different evaluations/reports e.g. in the estimation of the range of coverage, to be summarized individually.

The application "production levels" is divided into two areas:

- Production levels and their editing functions
A production level may be selected, added or edited in this dialog.
- Assignment functions and their relevant editing functions
This screen shows the machines and material buffers assigned to the selected production level.
Further machines and material buffers are also added or edited.

Selection criteria

The application provides the following selection criteria:

Production level

The production level may be selected or entered.

Material buffer

The [material buffer](#) can be selected or entered.

Machine

The machine may be selected or entered.

Field descriptions "production levels"

Production level

The production level may be selected or entered.

Description

Designation of the production level

Editing functions

The following dialog opens to edit a data record:

**Add**

Inserts a new production level

**Edit**

Edits an existing production level

**Delete**

Deletes the selected or several selected production levels

"Production levels assignment" detail application

The "assignment" detail application shows the below-mentioned data for each production level:

Category, type

These assignments are possible:

- Machine
- Material buffer

Name

Subject to the assignment, this field shows the machine number or the material buffer.

Description

Comment field / text field

Company

Abbreviated company name

Cost center

Affected cost center

Site / company

Affected site

Editor

Last editor of the data record

Last modification

Last modification to the data record.

The documentation dealing with the [assignment of production levels](#) describes the editing details for the selected production levels and, as a result, the detail application "assignment".

6 Closed Loops / Supply Relationships

Summary

Menu	Master data → Material → Supply relationships
Transaction code	intsc
Function authorization	intsc

Usage

By forming closed loops/supply relationships, the successor and predecessor levels of defined production levels can be defined. One production level can have several successor and predecessor levels.

Integration

The closed loops/supply relationships are used in the evaluation of ranges as well as in the e-kanban process as base data.

Requirements

The production levels must already be created in the system.

Selection criteria

The following selection criteria are available in the application:

Preceding level/supply - production level

Displays all supply relationships with the selected production level

Preceding level/supply - description

Displays all supply relationships with the selected description

Subsequent level/consumption - production level

Displays all supply relationships with the selected production level

Subsequent level/consumption - description

Displays all supply relationships with the selected description

The user authorization "kov" is required for displaying the below-mentioned fields.

Material number

Shows all closed loops/supply relationships matching the entered material number.

Closed loop/supply relationship ID

Shows the closed loop/supply relationship matching the entered ID that is specific within the system.

Field descriptions

"General" index tab:

Preceding level/supply

Selected preceding level/supply of a closed loop

If e-kanban is in use the preceding level/supply (place of production) represents the starting point of the closed loop for a kanban object. The supply is chosen from the production levels created in the system.

In case the supply is a supermarket, warehouse or intermediate production buffer, it has to be created as material buffer within MES and is chosen from the production levels (the material buffers are assigned to).



If e-kanban is in use, it is not possible to directly assign a machine group as the supply or a relevant material buffer for an entire machine group. But in fact the same material buffer is entered as common material buffer of the entire machine group for all affected machines pertaining to a machine group. This common material buffer is then selected as the supply.

Subsequent level/consumption of the closed loop

Selected subsequent level/consumption of a closed loop.

If e-kanban is in use the consumption represents the end point (place of consumption) of the closed loop for a kanban object. The consumption is chosen from a production level created in the system.

In case the consumption is a supermarket or intermediate production buffer, it has to be created as material buffer within MES and is chosen from the production levels (the material buffers are assigned to).



If e-kanban is in use, it is not possible to directly assign a machine group as the consumption or a relevant material buffer for an entire machine group. But in fact a common material buffer (supplying the whole machine group) is entered or selected as preceding machine buffer for all affected machines pertaining to a machine group.

The user authorization "kov" is required for displaying the below-mentioned fields.

Material number

Material number of the kanban article for which the closed loop applies. Several closed loops (n entries) can be defined for a material.

Closed loop/supply relationship ID

Specifically identifies the closed loop/supply relationship. A supply relationship can only be created once within the system.

"Kanban" index tab:**Number of KBN in circulation**

Total number of all kanban objects circulating within this closed loop.

Minimum stock level of empty kanbans (start "yellow" zone)

The minimum stock level of empty kanbans represents the upper stock limit of empty kanban containers that may be reached before, for example, production (replenishment) is triggered automatically and a changeable, planned kanban order is generated initially. Hence, the minimum stock level of empty kanbans represents the limit between the "green" and "yellow" zone (status lights at the terminal). The minimum stock level needs to be configured for the display in:

- Electronic Kanban Board (AIP)

Maximum stock level of empty kanbans (start "red" zone)

The maximum stock level of empty kanbans represents the lower stock limit of empty kanban containers that may be reached. If this limit is exceeded, the planned kanban order, for example, automatically becomes an unchangeable, fixed kanban order. Hence, the maximum stock level of empty kanbans represents the limit between the "yellow" and "red" zone (status lights at the terminal). The maximum stock level needs to be configured for the display in:

- Electronic Kanban Board (AIP)

Planned KBN quantity

Planned, defined quantity of kanban objects (contents).

7 Kanban Configuration

Summary

Menu	Master data → Material → Kanban configuration
Transaction code	kres
Function authorization	kres

Utilization

To be able to use e-kanban, each kanban included in the system is used as an object similar to a resource.

The master data for all used kanban objects are created and managed in this application. In HYDRA kanban objects are considered as resources with the pre-defined resource type **"KBN"**.

Selection criteria

The following selection criteria are available in the application:

Resource from ... to ...

This selection criterion refers to the unique ID of the kanban object. Wildcards (placeholders *) can be used.

Designation

Designation of the kanban object.

Kanban article

Article number/material number for which the kanban object is used.

Closed loop

Unique ID of the closed loop/supply relationship for which the kanban object was created.

Field descriptions

"General" category

Resource/ object

This field includes the ID matching the resource or kanban object to be entered. The ID/number is unique for each resource type.

Allowed characters are:

ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890/_.-+#. Blank-

Other special characters are not allowed. For technical reasons, * (asterisk) and % (percent) can be input, but are nonetheless not permitted because they are not valid characters. When the input field is exited, lower case letters are automatically transformed into CAPITAL LETTERS.

Designation

This field has been designed to assign a short, distinct designation to each kanban object. This designation is displayed in reports and overviews and it is useful for orientation.

Kanban article

Article number/material number for which the kanban object is used. This field is a mandatory field and key field. The article number/material number can also be chosen from the material master.

Closed loop/supply relationship

Unique ID of the closed loop/supply relationship for which the object was created. This field is a mandatory field and key field. The ID can also be chosen from the application "supply relationships".

Consec. number

Consecutive number of the kanban object from the total number of kanban cards (e.g. 1 out of 10 kanban objects) circulating in this closed loop/supply relationship.

"User fields" category

User fields offer the possibility to store further customer-specific information to MES besides the fields available in MOC standard. The tab provides eight sub tabs each of which providing eight user fields. The so called user field key determines which user fields are involved and which meaning they have.

Object type

User fields are configured in a fixed manner for the resource type = KBN

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next. User field keys are defined in coordination with the customer during the customizing process.

User fields

The following user fields are available after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6

Field data type	Number of fields
Text field, length 20	14
Text field, length 40	2

A maximum of 8 fields are shown for each page.

User field keys are not defined by default in the system. The system has to be customized accordingly to be able to support this kind of user fields.

"Comment" category

The Comment tab allows additional comments about the resource to be stored in six input fields.

Toolbar



Insert

Opens the dialog for adding a comment.



Copy

Opens the dialog for copying a comment.



Edit

Opens the dialog for editing a comment.



Delete

Deletes one or several resources.



Resource status

Opens the application to configure the resource status/kanban status.



Kanban overview

Opens the application "kanban overview"



Status change

Opens the dialog for changing the resource status/kanban status.

8 Closed Loop/Terminal Assignment

Summary

Menu	Master data → Material → Closed loop/terminal assignment
Transaction code	tla
Function authorization	tla

Utilization

This application can be used to assign the required closed loops/supply relationships for each supply (AIP of the supply) to be displayed in the electronic kanban board at the terminal.

Selection criteria

The following selection criteria are available in the application:

Terminal

Terminal number/ identification

Closed loop

Unique ID of the closed loop/supply relationship assigned to the terminal.

Position

Position of the closed loop/supply relationship to be displayed at the terminal

Field descriptions

Terminal

Terminal number/ identification

Closed loop

Unique ID of the closed loop/supply relationship assigned to the terminal.

Position

Position of the closed loop/supply relationship to be displayed at the terminal

9 Kanban Stock Levels

Summary

Menu	Material management → Kanban → Kanban stock
Transaction code	ksto
Function authorization	ksto

Utilization

This application shows the current kanban requirements. The user is informed about the following:

- Current status of empty kanban objects for a material within a special, closed loop compared to the allowed minimum and maximum stock level.
- Relative fill level of empty kanban objects for each material and closed loop.
- Current status of all empty kanban objects for a material (total of all empty kanban objects accumulated from all closed loops) compared to the accumulated, allowed minimum and maximum stock levels.
- Quantities for each material within a closed loop/supplier relationship.
- Quantities for each material (total)

This shows when the next kanban order is generated/should be generated as well as the current stock level of material or the current stock level of empty kanban objects.

It is displayed for each material and closed loop/supply relationship as single entry within the grid. In addition, grouping by material numbers is allowed. The below-mentioned values are accumulated in this case:

- KBN in circulation
- KBN stock: empty
- KBN stock: filled
- KBN stock: miscellaneous
- Current quantities
- Planned quantities

Selection criteria

The following selection criteria are available in the application:

Closed loop

Unique ID of the closed loop/supply relationship

Material

Kanban material

Material designation

Designation of the kanban material

Field descriptions**Closed loop**

Unique ID of the closed loop/supply relationship in which kanban objects are circulating.

Material

Kanban material/kanban article from the material master

Material designation

Designation of the kanban material/kanban article from master data

Number of KBN in circulation

Number of kanban objects of this material or article/resource type (total)

Minimum stock level of empty kanban

Minimum stock level of empty kanbans → limit value from master data

Maximum stock level of empty kanban

Maximum stock level of empty kanbans → limit value from master data

KBN stock: empty

Number of all kanban objects of this material or article assigned to the "empty" status

KBN stock: filled

Number of all kanban objects of this material or article assigned to the "filled" status

KBN stock: miscellaneous

Number of all kanban objects of this material or article assigned to another status (neither "empty" nor "filled")

Current quantities

Shows the current quantity (e.g. number of pieces/quantity) for a material/article. The calculation is based on the kanban stock level with the "filled" status and the actual quantity.

Planned quantities

Shows the planned quantity (e.g. number of pieces/quantity) for a material/article. The calculation is based on the kanban stock level with the "filled" status and the planned quantity.

Relative fill level

Shows the relative fill level for a material/article. The calculation is based on the relation between KBN with status = empty / initial and KBN in circulation. Data is entered in percent.

Toolbar



Generate kanban order

Authorization: kbn.generate

10 eKanban Data Collection

Main view

The main view provides the following functions for the kanban process. The available functions change, when the screen changes (supply or demand).

- Electronic kanban board
- Fill kanban
- Empty kanban

Electronic kanban board

On the supply side, the AIP displays the *Electronic kanban board*. The electronic kanban board informs the user about the number of empty kanban objects of a specified material in a defined control cycle.

To open the electronic kanban board, click the button *Electronic kanban board* on the main view. The system then loads all kanban items (including the relevant additional information) that are part of the control cycles assigned to this AIP.

The following information is displayed for the user:

- Status
 - Green:
The number of empty kanban objects is less than the minimum number of empty kanban objects
 - Yellow:
The number of empty kanban objects is greater than or equal to the minimum number of empty kanban objects
 - Red
The number of empty kanban objects is greater than or equal to the maximum number of empty kanban objects
- Material
- Control cycle
- Demand
- Kanban objects in the control cycle
- Maximum number of empty kanbans
- Minimum number of empty kanbans

Fill kanban (dialog KBN_FILL)

The dialog *Fill kanban* is used on the AIP terminal on the supply side. Here, you set the status of the actual kanban object in the system from *empty* to *filled* when the container has been filled. The dialog workflow is as follows:

- Open the dialog *Fill kanban* on the main view.
- Enter/scan the material number
- Select the control cycle for the material number entered
 - The system only displays the control cycles and the additional information if the control cycles are configured for the material number entered.
 - The following additional information is displayed for the control cycle:
 - Supply / demand
 - Objects in the control cycle
 - Full objects in the control cycle
 - Empty objects in the control cycle
 - Planned quantity per object
- The fields *Supply* and *Demand* are automatically filled when you select the control cycle.
- When you have selected the control cycle, the system automatically suggests the next kanban object with status *Empty*. If kanban objects with status *Initial* are available in the system, these objects are used first. The field *Consec. No.* is filled accordingly. Also the fields *Kanban quantity* and *Planned kanban qty.* are filled with the value of the planned kanban quantity stored in the resource master data. The user can change the quantity in field *Kanban quantity* and enter another actual quantity.
- After confirmation, the dialog KBN.FILL is sent to the server.
- A label/an accompanying material document is printed for this kanban object.



- Kanban containers need not be filled 100 %.
- The labels are destroyed when the material is used up (demand). The labels are only destroyed when the container has been emptied or the dialog KBN_EMPTY has been executed.

Empty kanban (dialog KBN_EMPTY)

On the demand side, the dialog *Empty kanban* is used on the AIP terminal. Here, you set the status of the actual kanban object in the system from *filled* to *empty* when the containers have been processed. When the dialog has been executed, the label/accompanying document for this kanban object is destroyed and the kanban object is assigned the status *Emptied*. The dialog workflow is as follows:

- Open the dialog *Empty kanban* on the main view.
- Enter/scan the kanban object
- The following information is automatically displayed:
 - Consec. no.
 - Material number
 - Kanban objects in the cycle
 - Control cycle ID
 - Supply
 - Demand
 - Planned kanban quantity
 - (Actual) kanban quantity
- The user confirms the dialog via staff badge number and clicks the button *Empty kanban*. The dialog is closed. The kanban object is set to the status *Empty* in the system. The kanban object is posted back to the supply side.
- If the user clicks the *Cancel* button, the dialog is interrupted and exited. The status is not changed and in the system, the kanban object is still on the demand side.



- The system only empties kanban containers completely (100%)
- For this reason, only kanbans/containers that have been emptied entirely are posted. If a container includes a remaining quantity, the container is considered as full container and put back on the shelf. The withdrawal from the container is not known to HYDRA. The quantity entered on the label or accompanying document is unchanged (full quantity). The quantity is not adjusted and a new label is not printed.



- The kanban object number must be unique in the system. This way, the number can be used and the number of the control cycle need not be entered additionally.

Order list

You call the order list using the dialog *Log operation on*. The order list shows the normal orders and the kanban orders/OPs.

The order list of a machine (supply) displays the following orders:

- Production orders/OPs (planned for this machine/machine group) → data record is "blue"
- Fixed kanban orders (of the supply/machine group from the generated order/OP) → data record is "lilac"

- Planned kanban orders (of the supply/machine group from the generated order/OP) → data record is "orange"

You can log on kanban orders using the dialog *Log operation on*.